

Program : Diploma in Renewable Energy	
Course Code : 6291B	Course Title: Electric Power Distribution & Protection
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- The objective of this course is to set a firm foundation on the fundamental concepts of Power System Generation, Transmission, Distribution and Protection.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic Electrical Engineering concepts		Fundamentals of Electrical & Electronics Engineering	II

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify the basic concepts and design considerations of distribution systems.	14	Applying
CO2	Explain the classification of distribution systems and substations.	16	Understanding
CO3	Summarize the principle of operation of circuit breakers and protective relays.	14	Understanding
CO4	Illustrate the protection schemes for electrical machineries and transmission lines.	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the basic concepts and design considerations of distribution systems.		
M1.01	Summarize the basic concepts of distribution system and distribution system planning.	2	Understanding
M1.02	Classify distribution systems on the basis of interconnections.	4	Understanding
M1.03	Classify different loads on power systems.	4	Understanding
M1.04	Identify the design considerations of various distribution feeders.	4	Applying
Contents: General Concepts: Introduction to distribution system, Distribution system planning, Factors affecting the Distribution system planning. Connection scheme & Requirements of Distribution system- Radial system-Ring main system-Interconnect system-proper voltage-availability of proper voltage-reliability Loads on power system: Load growth, Classification of loads (Residential, commercial, Agricultural and Industrial) and their characteristics- Relationship between the load factor and loss factor.. Distribution Feeders: Design Considerations of Distribution Feeders: Radial, loop and network types of primary feeders.			
CO2	Explain the classification of distribution systems and substations.		
M2.01	Summarize different classifications of distribution systems.	4	understanding

M2.02	Explain the basic principles of the DC distribution system.	4	understanding
M2.03	Classify substations based on the service requirements and constructional features.	4	understanding
M2.04	Illustrate various substation equipment symbols and bus bar arrangement.	4	Understanding
	Series Test – I	1	

Contents:

Distribution system: Classification of distribution system – According to nature of current, Type of construction, Scheme of connection- AC distribution system-primary and secondary distribution system

Dc distribution system: OH & UG distribution system-connection scheme -2 wire & 3 wire distribution system-Connection scheme –Radial system, Ring main system, Interconnect system

Substation: classification of substation- According to service requirement & constructional features- Indoor & outdoor substations- Transformer substation- Pole mounted substation- underground substation.

Substation equipments: Symbols & purpose of various Equipment in indoor & outdoor substation- Bus bar arrangement in substation.

CO3	Summarize the principle of operation of circuit breakers and protective relays.		
M3.01	Explain the principle of operation of circuit breakers.	4	understanding
M3.02	Illustrate different types of circuit breakers.	4	understanding
M3.03	Summarize the basic concepts of protective relays.	2	understanding
M3.04	Explain the classification of protective relays.	4	understanding

Contents :

Circuit breakers – principle of operation- formation of arc- Arc quenching theory, Restriking Voltage-Recovery voltage, RRRV. Interruption of Capacitive currents and current chopping (Brief Description Only).

Types of Circuit Breakers: Construction- principle of operation- Air blast CB – Oil CB – SF6 CB – Vacuum CB – CB ratings.

Protective relays- Basic concepts- Essential qualities- Time current characteristics.

Classification of relays- Construction- principle of operation- merits & demerits- Attracted armature type- Induction type (non-directional & directional)- static relays

CO4	Illustrate the protection schemes for electrical machineries and transmission lines.		
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M4.01	Illustrate various schemes for the protection of alternators.	3	understanding
M4.02	Summarize different protective schemes for transformers.	3	understanding
M4.03	Explain various schemes for the protection of transmission lines.	4	understanding
M4.04	Summarize the mechanism of lightning and its protection.	4	understanding
	Series Test – II	1	

Contents :

Protection of alternators: Earth fault Protection and Differential protection.

Protection of transformers- Differential Protection-Buchholz Relay

Protection of transmission lines: schematic arrangement and principle of time graded over current protection - differential pilot wire protection-distance protection.

Protection against Lightning-definition-mechanism of lightning discharge- -harmful effects of lightning.-protection against lightning-devices used- overhead ground wires- Lightning arrestors-horn gap arresters(basic diagram and working only)

Text / Reference

T/R	Book Title/Author
T1	“Electric Power Distribution system, Engineering” – by TuranGonen, McGraw–hill Book Company.
T2	“Principles of Power Systems”- V. K. Mehta and Rohit Mehta- S. Chand publications
T3	J.B. Gupta, “A course in Electrical Power”, Katharia and sons, 2004.
R1	Electric Power Distribution – by A.S. Pabla, Tata McGraw–hill Publishing company

Online Resources

Sl.No	Website Link
1	https://www.electrical4u.com/protection-system-in-power-system/
2	https://nptel.ac.in/courses/108101039
3	https://onlinecourses.nptel.ac.in/noc21_ee110/preview